VASILIKI ISMIROGLOU

PhD Fellow

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EDUCATION

2024-	**PhD in Computer Vision** Focus on Underwater Vision for Environmental Monitoring *Visual Analysis and Perception Laboratory, Aalborg University, Denmark*
2023-2024	**MSc in Computer Engineering** (partially completed) Specialization in AI, Vision and Sound *Aalborg University, Denmark*
2015-2022	**Integrated MSc in Applied Mathematical and Physical Sciences** Specialization in Theoretical Physics *National Technical University of Athens, Greece*

PROFESSIONAL EXPERIENCE

2023-2024	**Artificial Intelligence Engineer** Student Worker *TeeJet Technologies, Denmark*

RESEARCH INTERESTS

- The performance of deep learning models in turbid, low-visibility underwater environments and ways to improve it.
- Marine optics and physics-grounded approaches to synthetic underwater data generation and scene rendering.
- Image information content and ways to quantify it.

ADMINISTRATIVE EXPERIENCE

Workshop Coorganizer

- Workshop on Marine Environmental Monitoring and Assessment using AI (MEMAAI 2025).

Reviewer

- The IEEE/CVF Conference on Computer Vision and Pattern Recognition
- International Conference on Computer Vision
- International Journal of Computer Vision
- Scandinavian Conference on Image Analysis

TEACHING EXPERIENCE

Teaching Assistant

Fall 2024 & Fall 2025	**Image Processing and Computer Vision** Computer Engineering – AI, Vision and Sound MSc 1st Semester *Aalborg University, Denmark*
Fall 2024 & Fall 2025	**Advanced Robotic Perception** Robotics MSc 1st Semester *Aalborg University, Denmark*

Group Project Supervisor

Fall 2024	**Underwater Image Enhancement Utilizing Dehazing and Color Correction** Robotics BSc 3rd Semester *Aalborg University, Denmark*
Fall 2025	**Leveraging Pose Estimation and Tail Features for Experimental Autoimmune Encephalomyelitis Estimation in Mouse Models** Computer Engineering – AI, Vision and Sound MSc 1st Semester *Aalborg University, Denmark*

PUBLICATIONS

Peer-reviewed Papers

- *Bengtson, S. H., Lehotský, D., Ismiroglou, V., Madsen, N., Moeslund, T. B., & Pedersen, M. (2025).* ***AutoFish: Dataset and Benchmark for Fine-grained Analysis of Fish.*** *In Proceedings of the Winter Conference on Applications of Computer Vision (pp. 1598-1607).*
- *Fredenslund, L. L., Ismiroglou, V., Moeslund, T., & Nasrollahi, K. (2025, July).* ***Privacy Aware Human-Object Interaction in the Wild Novel Dataset.*** *In International Conference on Advanced Concepts for Intelligent Vision Systems (pp. 121-133). Cham: Springer Nature Switzerland.*
- *Ismiroglou, V., Pedersen, M., Bengtson, S. H., Aakerberg, A., & Moeslund, T. B. (2025, September).* ***Sea-ing Through Scattered Rays: Revisiting the Image Formation Model for Realistic Underwater Image Generation.*** *In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops.*

Peer-reviewed Abstracts:

- *Ismiroglou, V., Pedersen, M., Bengtson, S. H., Aakerberg, A., & Moeslund, T. B. (2025, November).* ***Generating Realistic Underwater Images using a Revised Image Formation Model.*** *AI in Science Summit 2025.*

Other:

- *Ismiroglou, V., Pedersen, M., Bengtson, S. H., & Moeslund, T. B. (2025).* ***Inside Out … Turbid Waters, Clear Sight: Can AI See What We Cannot?*** *Journal of Ocean Technology, 20(2), 146-147.*